

OUMAR BARRY

Software Architect | AI Products, Platforms & Infrastructure

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SUMMARY

Software Architect with ten years of hands-on experience building AI platforms, developer tools, digital products and production infrastructure across telecom, finance, government and SaaS. Owns work from product and architecture through implementation, deployment and operations. Current work includes a nine-service AI platform, self-hosted cloud and ISP systems, payment infrastructure and open-source tools in the Nuxt and Python ecosystems.

PROFESSIONAL EXPERIENCE

Executable Technologies, DBA Apek Group *remote*

Software Architect *October 2022 - present*

Earlier roles: Senior Software Engineer, February 2020 - October 2022; Software Engineer, July 2016 - February 2020

- Co-architected and continue to build o3ai with Amadou, a nine-service, multi-product AI platform evolved from nearly two years of work on Freddy and ORIA. Freddy was built on a corpus of 100,000+ African legal and case-law documents.
- Built and continue to harden a self-hosted, identity-aware speaker gate that creates an ephemeral voiceprint and filters non-matching speech before model VAD for noisy and shared-space conversations.
- Architected and operate ExeApps' self-hosted platform for application delivery, object storage, source control, CI/CD, observability and backups using Proxmox, Dokploy, GitLab and open-source infrastructure.
- Built BIETRIE for a 20,000-attendee concert. It later supported events of 60,000 to 80,000 attendees before being reworked as a multi-tenant product.
- Led architecture and delivery across digital public services, telecom and field operations, including Poste Smart, SIGUI and Biome.exe, an offline biometric system delivered end to end in under 30 days.
- Earlier work includes leading a five-engineer hospital platform and building a mobile shop-management product used by 600+ stores in Conakry.

Selected partner engagement through ExeApps

Leader Net Services Guinea, West Africa

Technology Consultant *June 2024 - present*

- Lead software, network and infrastructure work for a Guinean ISP, coordinating 10+ engineers and remaining hands-on during critical service migrations and production incidents.
- Built ISPMAN to unify billing, QoS, network, customer and field operations. With Amadou, added local LibreQoS analytics using self-hosted ClickHouse.
- Built and deployed EPL connectivity from Conakry to Telehouse TH2 in Paris, a self-hosted Open5GS mobile core for private LTE, and BGP peering between Guinea and Sierra Leone.
- Deployed dedicated QoS for a major mining client. Recurring bandwidth complaints dropped to zero after rollout.

OPEN SOURCE

- Created better-auth-py, a framework-agnostic Python port of Better Auth with FastAPI, Litestar, Flask and Django integrations, 35 social providers and 26 plugins.
- Built nitro-openapi-schemas, a working FastAPI-style prototype for schema-driven OpenAPI 3.1 in Nitro/H3, and authored 20+ starters for the Nuxt and Nitro ecosystems.
- Current open-source work also includes fastapi-file-routing and Mongoz, alongside contributions to Nuxt, Nitro, UnoCSS, TresJS, nuxt-modules/ionic and web3-onboard.

SELECTED PRODUCT WORK

- **Pygmalion:** Nuxt-native commerce framework with 282 API endpoints, 98 database tables and 790 tests. npm release pending.
- **Horizon:** Rust and PostgreSQL payment switch that sustained 1,753 settled TPS at 33 ms internal p99 in a local benchmark across 315,522 transfers, with no non-business errors.
- **o3money:** Digital wallet and public-finance platform built under an active Visa Global partnership. Technical integration with Visa's platform is in progress.
- **SuperBudget:** Complete Flutter prototype for voice-first expense capture, structured review and financial guidance. Currently evaluating models that can run locally on phones.
- **Ozone:** Account-based Layer 1 built from scratch in Rust, with a Flutter wallet and Nuxt explorer. The current implementation is single-node.
- **Other active products:** GRIOT, Afromaps, pointgn, o3health and an African-folklore Metroidvania.

SKILLS

Languages: TypeScript, Python, Rust, Java, Kotlin, C, Swift

AI and data: FastAPI, RAG, real-time voice, Qdrant, Neo4j, PostgreSQL, ClickHouse

Product engineering: Nuxt, Vue, React, Next.js, Nitro, H3, Node.js, Flutter

Infrastructure: Cloudflare Workers, Docker, Proxmox, GitLab CI/CD, observability, object storage

Networks: BGP, private LTE, QoS, traffic analytics, ISP operations

EDUCATION

Licence MIAE, Computer Science and Business Management - *Université Nongo Conakry, 2017 - 2021*